UNIVERSITY OF CAPE TOWN
ECO4102Z/ECO4122Z
Class Example

1. Mr. Bheki (hereafter "Mr. B") is a single 50-year-old, hard-charging pharmaceutical executive earning R2,500,000 per year. He lives extravagantly (self-control) and occasionally overspends, but he has saved approximately R1.5 million. Mr. B had a mild heart attack last year but now has a clean bill of health. His primary financial goal is to retire comfortably at 65 and to donate R3 million to his alma mater (he cannot obtain adequate life insurance to cover the gift). You have been working with Mr. B for less than a year. You've drafted a financial plan but have yet to modify Mr. B's preexisting allocation (nearly 100 percent equities). However, you have developed a good working relationship with Mr. B, who listens intently and seems receptive to your recommendations. You believe that Mr. B is a wellgrounded person and is self-aware, but you also believe that he has some behavioral issues to deal with.

   Analysis

   At the outset of your relationship, you administer a risk tolerance questionnaire which outlines a more conservative, mean-variance optimized allocation. However, you are worried that Mr. B may not fully buy into the idea. Your concern is that a severe downward market fluctuation may cut into Mr. B's daily living expenses, including possible health expenses.

   Your financial planning software tells you that with a less aggressive portfolio, Mr. B can still meet his primary financial objectives. However, Mr. B thinks that he is a (overconf) very good investor, and you are worried that such a change might cause him to (regret-aversion) regret not being more aggressive. He agrees to complete a comprehensive behavioral bias questionnaire, and his results show susceptibility to:

   - **Overconfidence bias :** (cog) the tendency to overestimate one's investment savvy.
   - **Regret aversion bias:** (emo) the tendency to avoid making a decision for fear that the decision may cause regret later on.
   - **Self-control bias:** (cog x emo) the tendency to spend today rather than save for tomorrow.

   The mean-variance optimized allocation you initially calculated for Mr. B was 70 percent stocks, 25 percent bonds, 5 percent cash–Mr. B's risk tolerance better suits a more balanced portfolio. You also have obtained confirmation of the specific behavioural biases that are probably causing the distortion in Mr. B's portfolio. Your job is now to answer these questions:

   (a) What effect do Mr. B's biases have on the asset allocation decision?

   (b) Should you moderate or adapt to his biases?

   (c) What is the best practical allocation for Mr. B?

a) overconfidence bias:
- May cause him to be more risky + overweight equity & underweights bonds / cash

regret-aversion bias:
- may lead him to stay with a risky portfolio to avoid regretting not being more aggressive -missing out on surge

self-control bias:
- He lives extravagantly & occasionally overspends. For this reason, he may want an aggressive portfolio to fund his lifestyle - fixed income need to supplement lifestyle
-Allocation towards equities

b) • He is of high wealth, low SLR
• His biases are cognitive with 1 being emo ↗ emo

∴ Moderate & adapt ∴ adapt
Advise him to keep comfortable cash reserve

c) Equity : 50 to accomodate conservative risk tol 70 → 75
Bonds : 45 to accomodate / supplement lifestyle 25 → 15
Cash : 5 5 → 10

should allocate adequate living funds even if mkt turns sour